

GIRIDHAR R

rgiridhar1212@gmail.com | 7695949443 | linkedin.com/in/rgiridhar12 | github.com/giridhar122 | Portfolio | Bengaluru

PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience developing and deploying AI-integrated solutions to solve real-world problems. Proven ability to build optimized machine learning models and collaborate effectively in fast-paced environments to deliver high-quality software solutions.

EDUCATION

Amrita Vishwa Vidyapeetham, Bachelor of Technology in Computer Science and Engineering
CGPA:7.56/10

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering.

PROJECTS

Formation Energy prediction for Bimetallic Nanoparticle,

Python, PyTorch, PyTorch Geometric, Graph Convolutional Networks (GCN), Graph Neural Networks, NumPy | 02/2026

- Developed a Graph Convolutional Network (GCN) using PyTorch Geometric to predict formation energy of Cobalt-Gold (CoAu) nanoparticles from 3D atomic coordinates.
- Processed nanocluster structures containing 586 atoms per structure for memory efficient learning.
- Trained and optimized the model using Adam optimizer and Mean Squared Error (MSE) loss, achieving stable convergence and reducing inference time to under 100 ms compared to traditional DFT simulations.
- Implemented validation strategies, including Early Stopping, to prevent overfitting and enhance model stability during training.

Community Driven Sustainable Development Using Data-Driven Solutions,

Data Analysis, Participatory Rural Appraisal (PRA), Stakeholder Engagement, Statistical Analysis | 05/2025

- Collected, organized, and analyzed qualitative and quantitative data from 50+ community members using Participatory Rural Appraisal (PRA) techniques to identify local infrastructure and resource challenges.
- Interpreted field data to uncover critical issues, including 60% of households lacking piped water and 96% without waste management, supporting evidence-based decision making and sustainable solution planning.
- Collaborated with multidisciplinary teams to evaluate community requirements and propose data-driven, technology-enabled solutions for water management, agriculture, and waste management.

Smart Accident Detection & Notification System,

Python, TensorFlow/Keras, OpenCV, Convolutional Neural Networks (CNN), Twilio API, Geopy | 01/2024

- Developed and trained a scalable Computer Vision architecture utilising Convolutional Neural Networks (CNNs) for real-time video frame analysis to detect traffic accidents with over 90% accuracy.
- Applied OpenCV for image and video preprocessing, enabling continuous frame analysis and improving model performance under varying environmental conditions.
- Utilized Twilio and Geopy to trigger automated geo-tagged notifications within 5–7 seconds, optimizing emergency response times.

SKILLS

Programming and Core Concepts: Python, HTML, SQL, Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Management Systems (DBMS), Computer Networks.

Tools: Git, GitHub, VS Code, MySQL, Jupyter Notebook, Google Colab.

Machine Learning and Computer Vision: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, OpenCV, NumPy, Matplotlib.

Soft Skills: Communication and Collaboration, Analytical Thinking, Problem Solving.

ACHIEVEMENTS AND PUBLICATIONS

Road Safety Hackathon, IIT Roorkee: Achieved **2nd position** in the Accident Prediction & Prevention category. (01/2024)

Mython Data Hackathon, IIIT Sricity: Achieved **3rd place** in a data analytics hackathon. (02/2024)

HacksDay Hackathon, Oxdays: Worked on real-world problem statements and contributed to solution refinement. (12/2024)

IndustriAI Hackathon, IIT Madras: Participated and attended workshops during Shaastra 2025. (02/2025)

Live in Labs, Amrita Vishwa Vidyapeetham, Chennai: Field Research Participant. (05/2025)

Publication: Community-Driven Sustainable Development Through Human-Centred Design and Participatory Rural Assessment, Co-author | SSRN (Elsevier) | 11/2025